

Voronoi diagrams



1

- a All the points in cell A are closer to site A than the other sites.
- b These points are equidistant between site A and site B.
- c This is the only point that is equidistant to all three sites.

2

- a Site A because the point lies in cell A.
- b No solution provided.

c

- i Site B
- ii Site A
- iii Site C
- iv Site B

3

- a No solution provided.
- b No solution provided.
- c No solution provided.
- d No solution provided.
- e No, because there is no point that intersects with all the edges that divide the four cells.

Applications

4

- a School D

b

- i School A
- ii 2.86 km

c

- i Either school C or D.
- ii School A

5 Example answer:

The point (10, 50) would be a good location as it is quite far from most of the towns and is on the edge between Town E and J, so it will be equidistant from both towns so not too close to either town.